12/10/24

Activity - 1

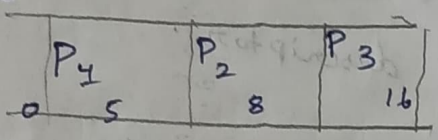
Process Scheduling

4) Three processes arrive at time 0:

Process	Burst Time
P1	5
P2	3
P3	8

Using FCFS, the average waiting time is:-

Ready Queue



TAT = CT - AT
WT = TAT - BT

PROCESS	BT	AT	CT	TAT	WT
P1	5	0	5	5	0
P2	3	0	8	8	5
P3	8	0	16	16	8

Average waiting time

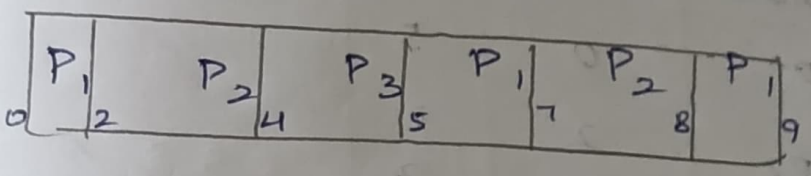
$$= \frac{5+8}{3} = 4.33$$

Option b) 4.33

2) Consider Round Roubin with time quantum = 2

Procers	Burst
P1	5
P2	3
P3	1

All arrive at time 0. which procers completes first?

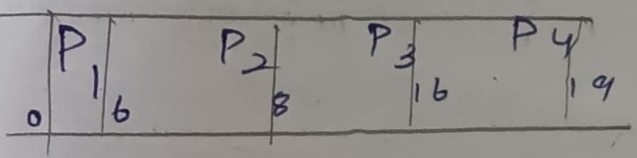


The answer:-

(C) P_3 completes first.

3) Consider the following procers scheduled using FCFS. All processes arrive at time 0.

Procers	burst Time	AT
P1	6 ms	0
P2	2 ms	0
P3	8 ms	0
P4	3 ms	0

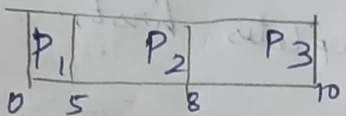


$P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4$

Process	BT	AT	CT	TAT	WT
P1	6	0	6	6	0
P2	2	0	8	8	6
P3	8	0	16	16	8
P4	3	0	19	19	16

b)

Process	AT	BT
P1	0	5
P2	2	3
P3	4	2



Process	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	2	3	8	6	3
P3	4	2	10	6	4

Answer:- Avg TAT $= \frac{5+6+6}{3}$

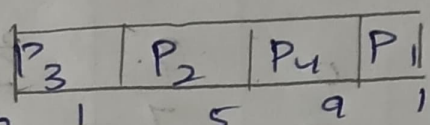
$= 5.67$

Shortest Job first

Process	BT
P_1	7ms
P_2	4ms
P_3	1 ms
P_4	4 ms

By SJF,

$\Rightarrow P_3 \rightarrow P_2 \rightarrow P_4 \rightarrow P_1$



TAT - BT = WT

TAT = CT - AT

Process	BT	AT	CT	WT	TAT
P_1	7	0	16	9	16
P_2	4	0	5	1	5
P_3	1	0	1	0	1
P_4	4	0	9	5	9

Avg Waiting Time

$$= \frac{9+1+5}{4}$$

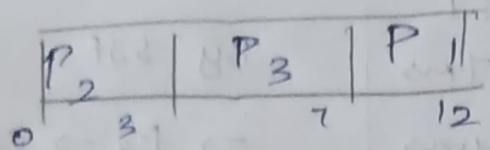
$$= 3.75$$

Avg WT = 3.75

5)

Process	BT	Priority	AT
P1	5	3	0
P2	3	1	0
P3	4	2	0

$P_2 \rightarrow P_3 \rightarrow P_1$

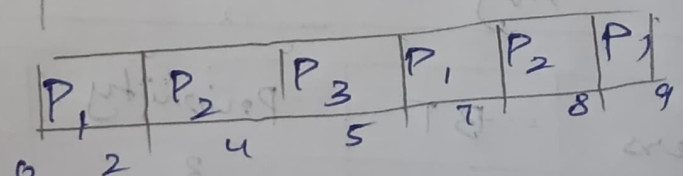


Process	BT	AT	CT	TAT	WT
P1	5	0	12	12	7
P2	3	0	3	3	0
P3	4	0	7	7	3

$$\text{Avg WT} = \frac{7+3}{3} = 3.33$$

6) Time Quantum = 2 ms

Process	BT	AT	CT	TAT	WT
P1	5	0	9	9	4
P2	3	0	8	8	5
P3	1	0	5	5	4



The completion time of $P_2 = 8$

7) Time Quantum = 3ms

Process	BT	AT	CT	TAT	WT
P_1	4	0	9	9	5
P_2	5	0	11	11	6
P_3	2	0	8	8	6

